

Benjamin Abensur

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EDUCATION

EPFL — Swiss Federal Institute of Technology in Lausanne

Expected Graduation: 2028

Bachelor of Science in Communication Systems

- Relevant Coursework: Introduction to Machine Learning (5.75/6), Algorithms I (5.5/6), Linear Algebra (5.5/6), Probability & Statistics, Numerical Methods for Visual Computing & ML, Computer Architecture, Computer Systems, Responsible Software

WORK EXPERIENCE

Biosynex, Incoming AI Engineer (part-time contractor); Remote (Euronext-listed rapid diagnostics)

Starting Oct 2026

- Engaged to introduce the executive team to applied AI, then build AI tools into internal workflows.

EPFL, Teaching Assistant, Mechanics (PHYS-101) and Linear Algebra (MATH-111)

Sep 2026 – Present

- Selected on academic merit for a Mechanics course of **1,000+** students (CS, Chemistry, EE); lead a weekly **2-hour** exercise session and answer questions on the course's Ed forum.

BABOO, Founder & Event Producer; Lausanne, Switzerland

2025

- Produced **3** club nights end to end; one night at Noche Club (May 2025) brought in **CHF 5,000** with an internationally touring DJ.

NanoSynex (Technion medtech spin-off), AI & Automation, informal, reporting to the CEO; Remote

2024 (3 months)

- Deployed two offline LLMs (Ollama; Mistral 7B, Llama 3 8B) on the CEO's Macs behind a router with a shared Obsidian memory, so confidential R&D and investor documents never left the machine (GDPR).
- Built Python scripts triaging the CEO's Gmail and Outlook inboxes with an LLM classifier, before native AI email integrations existed.
- Harmonised and cleaned the test reader's experiment exports into one analysis-ready dataset (Python, pandas), standardising antibiotic names and flagging incomplete or duplicate runs.

Sneaker Resale Venture, Partner (team of 4)

2019 – 2021

- Secured a license to Flare AIO, a sell-out EU sneaker bot, by automating its restock purchase; ran it on Foot Locker and Snipes drops.
- Resold the license for about twice its cost; generated roughly **€10k** over the period.

PROJECTS

Gaming Addiction Prediction — ML from scratch, CS-233 (team of 3)

github.com/Benabens/addiction-classifier-numpy

Python, NumPy, no ML libraries

- Predicted gaming-addiction level (3 classes) and score (0–10) from gaming and mental-health data with KNN, logistic/linear regression, K-Means and an MLP with hand-written backpropagation; two written reports.
- Inverse-frequency weighted cross-entropy lifted rarest-class recall from **0 to 0.69** and macro-F1 from **0.55 to 0.76** (5-fold CV).

ReCHor — public-transport journey planner, CS-108

github.com/Benabens/ReCHor

Java, JavaFX

- Finds every Pareto-optimal journey between two Swiss stops (Connection Scan Algorithm) over real CFF timetables (~**258 MB**).
- Memory-mapped, bit-packed timetable storage; JavaFX UI with autocomplete and iCalendar export; **44** classes, ~**3,600** lines of tests.

ICoop — two-player cooperative 2D game, CS-107

github.com/Benabens/ICoop

Java

- Fire-and-water co-op game with a boss fight; interactions via double dispatch (visitor pattern). ~3,700 lines.

Kairo — AI career CRM, with Trame as its networking module

Next.js, TypeScript, Python, PostgreSQL (Supabase, row-level security), Claude API, GitHub Actions, Vercel

- Designed and shipped a daily pipeline aggregating **16,000+** internships, jobs and programmes from **35** career boards, Indeed/LinkedIn (30 markets) and hackathon feeds, each scored 0–100 by rules + an LLM.
- LLM extraction of deadlines and eligibility, each answer backed by a verbatim quote; ~37k lines, **160+** tests, CI, on Vercel.
- Trame*: PWA turning freeform notes into structured contacts via Claude, linked to Kairo to show who I know at each employer.

Cortex — AI exam-preparation platform

github.com/Benabens/cortex

Next.js, TypeScript, PostgreSQL, Drizzle, Claude API, LaTeX, Railway

- Learns each course's exam format from past papers, weights every topic by how heavily it was examined, schedules spaced revision on the student's weak points, and generates faithful practice exams via a multi-pass LLM pipeline.
- Answers accepted only if verified by sandboxed code execution or symbolic checks; per-user Postgres schemas, cost caps; ~28k lines.

Other Projects: Verilog digital-logic library (42/42 self-checking testbenches); networking and UNIX-filesystem project in C (CS-202, in a pair); *VeriReason*, a PyTorch Transformer from first principles (in progress); Reed-Solomon QR-code generator (Java, CS-107); interactive mechanics simulators for my students.

LEADERSHIP & COMMUNITY INVOLVEMENT

Scout Summer Camp, Co-Camp Director; Mévouillon, France

Jul 2025

- Co-directed a one-month camp for **100** children from two local groups (Nice, Strasbourg); led a team of **20** counsellors, promoted team leads and wrote the camp's full educational project.
- Co-managed the **€80k** camp budget, running the €18k food line myself (local farmers, Intermarché); built the site at pre-camp: tents, dry toilets, wooden tables; organised outings and explorations.

EPFL Linear Algebra Bootcamp (MATH-111), Co-instructor; Lausanne, Switzerland

Nov 2025

- Gave the lectures and wrote the exercise sets of a paid one-week bootcamp for **30** first-year students (mid-semester break).

SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL, Verilog, x86/RISC-V assembly

ML & AI: PyTorch, NumPy, pandas, matplotlib, Claude API, Ollama

Web & Systems: Next.js, React, Node.js, PostgreSQL, Supabase, web scraping, GitHub Actions, Linux, POSIX sockets, Git

Spoken: French (native), English (fluent), Hebrew (basic), Spanish (basic)